

7	(a) wavelength associated with a particle that is moving	M1 A1	[2]
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	(b) (i) kinetic energy = $1.6 \times 10^{-19} \times 4700$ $= 7.52 \times 10^{-16} \text{ J}$ <i>either energy = $p^2/2m$ or $E_K = \frac{1}{2}mv^2$ and $p = mv$</i> $p = \sqrt{(7.52 \times 10^{-16} \times 2 \times 9.1 \times 10^{-31})}$ $= 3.7 \times 10^{-23} \text{ N s}$ $\lambda = h/p$ $= (6.63 \times 10^{-34}) / (3.7 \times 10^{-23})$ $= 1.8 \times 10^{-11} \text{ m}$	C1 C1 C1 C1	[5]
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	(ii) wavelength is about separation of atoms can be used in (electron) diffraction	B1 B1	[2]
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8	(a) (i) $x = 2$	A1	[1]
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